How solar panels work

A solar panel is a grid of photovoltaic cells made from silicon. When sunlight hits a cell, it knocks electrons loose, and the cell's structure pushes them in one direction. That flow of electrons is an electric current.

The current from the panels is direct current. An inverter turns it into the alternating current that your home appliances use. Extra power can go back to the grid, and many utilities credit you for it through net metering.

Panels still work on cloudy days but produce less. In the northern hemisphere, a roof that faces south and is tilted close to your latitude gives the most energy, and shade from trees or chimneys can cut output sharply.

A home system often pays for itself in six to ten years and then produces nearly free electricity for decades. Maintenance is mostly an occasional rinse and a check on the inverter.